

1. What is Node.js?

- A **JavaScript runtime environment**.
 - Allows running JavaScript **outside the browser** (e.g., on servers).
 - **Built on Chrome's V8 JavaScript Engine**.
 - Enables building fast, scalable network applications.
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2. Who Created Node.js and Why?

- **Created by:** Ryan Dahl
 - **Released in:** 2009 (initial work started in 2007)
 - **Reason:**
 - Traditional servers like **Apache** handled concurrent requests inefficiently.
 - Node.js was designed for **non-blocking**, event-driven, real-time applications.
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3. Installing Node.js

- Download from <https://nodejs.org>
 - Choose:
 - **LTS (Long Term Support):** Stable version recommended for most users.
 - **Current:** Latest features but less stable.
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4. Running JavaScript Files with Node

- Use the terminal/command prompt:

```
node filename.js
```

- Node provides its own runtime environment with built-in **APIs** like:
 - `fs` (file system)
 - `http` (server creation)
 - `path`, etc.
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5. Packages in Node.js

- **Packages** are reusable libraries or tools.
- Installed using **npm (Node Package Manager)**.
- Example:

```
npm install cat-me
```

6. Packages vs Modules

Feature	Package	Module
Definition	Third-party tools/libraries	Built-in features in Node.js
Source	Installed via <code>npm</code>	Comes with Node.js
Examples	<code>express</code> , <code>cat-me</code>	<code>http</code> , <code>fs</code> , <code>path</code>

7. Installing and Using a Package (e.g., cat-me)

- **Installation:**

```
npm install cat-me
```

- **Usage** in your code:

```
const catMe = require("cat-me");  
console.log(catMe());
```

8. HTTP Module in Node.js

- Built-in module to **create servers**.
- No installation required.
- Enables handling **HTTP requests and responses**.

9. Creating a Simple Server with HTTP Module

```
const http = require('http');

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js server!");
});

server.listen(3000, () => {
  console.log("Server is running on port 3000");
});
```